

# Schematic completion status

PetSafe 100-1339 R03 A | v0.9.30 | 2026-10-09

All catalog entries are drawn on one sheet. The remaining questions are listed explicitly below. This is a partially verified reconstruction, not a completed hardware validation or a routed PCB.

| Coverage                                                         | Open connections                                                   | Values / packages                                                                    |
|------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 149 entries; 1 A2 sheet; 22 stock + 4 custom symbol definitions. | 0 populated-entry pads + 0 DNP pads. 0 open PIC pads in the model. | 44 ceramic values unknown; 0 magnetic readings missing; 148/149 footprints assigned. |

## Power rails - current names

| Name     | Known path                                                                               |
|----------|------------------------------------------------------------------------------------------|
| BATTERY+ | Battery positive -> Q1 single pad/native3 (candidate drain); TP14/R43.2 on the raw side. |
| VSYS     | Q1.L/native2 -> U5/MX512H pin4; L1.1 and L2.1 supply sides.                              |
| VDD      | TP103, PIC20/C32, U5 pin1/VCC, V064/V063/V075/V082. D3.R via V082 is user-confirmed.     |
| VREF     | TP104, PIC4/RA2/V035/C39, R21.2/V071. U3.5 and R22 D/left/model2 at 1 ohm.               |
| GND      | Battery negative and ground copper.                                                      |

L1/L2 share the VSYS supply side. Their opposite ends are not shorted together. H\_RF\_VDD, H\_LDO\_IN, H\_RX\_VDD and H\_PIR\_VDD describe local branches; these are not additional confirmed independent supplies. Rail voltages have not been measured. L1 = 1.4 uH and L2 = 2.2 uH are user LCR readings; test frequency and isolation condition are not yet reported.

## What remains to finish

- Resolve the remaining functional paths:** One complete Q3 tuning cell; remaining antenna continuation and local-only PIC nodes; receiver/PIR hypotheses. TP3/R6/R7/D1 excitation routing is now photo-supported; timing remains unmeasured.
- Verify candidate devices and operating behavior:** Q1/Q8 and tuning-transistor pin roles, D2 identity, D6 breakdown/ICSP compatibility, S1 contacts and actual rail/gate voltages.
- Recover the required capacitor values:** 44 fitted ceramic values remain unknown. C5 is 470 uF / 16 V; L1 1.4 uH and L2 2.2 uH readings are already recorded.
- Finish symbols and physical assignments:** LED1 footprint and red/green channel mapping; motor lead orientation. C5 lead pitch and S1 pad fit/3D height remain qualified.
- Document optional and later reproduction work:** Empty option values/roles, J6 onward role, actual layer stack, daughterboard internals, firmware and a routed PCB.

## Validation and limits

- 83 modeled net partitions match KiCad 10.0.5. ERC retains 6 findings: 0 open pins, 0 isolated labels, 5 undriven power checks and 1 regulator-option output conflict. U6A is DNP; its common output with fitted U6 is retained and documented, not suppressed. No artificial NC/power flags hide gaps. Original photo hashes are unchanged. This proves file consistency, not all physical connections.
- Isolated modeled nodes: None. A PIC pin can have a local net and still lack an onward destination.

## Completion milestones

- Topology complete:** Populated components and functional block interfaces have supported connections; candidate functions/polarities are compatible and remaining exceptions individually documented.
- Value/package complete:** Required values and geometry recovered; exact manufacturer/ratings separated from equivalent design choices. Routed PCB remains a separate deliverable.

# Supply and device gaps

## E01 - D2 / marking 4P

**Known:** Marking 4P. Diode readings support R -> S -> L series-junction behavior in circuit. Photo R=upper-left, L=upper-right, S=single lower. User confirms R-C20 lower 1 ohm; L-ANT2/R46 upper; S-R46 lower. R46 unpopulated. S-ANT2 OL; S-GND 700 kohm, S-VREF 1.32 Mohm. Outer pads show high resistance to VDD/GND.

**Uncertain:** All three D2 local destinations are resolved. Exact part, ratings and physical package numbering remain unknown; internal series topology is probable from in-circuit tests. Other RX\_A\_PLUS branches remain inferred. The purpose/value of the empty R46 option is unknown; no resistor is fitted.

**Next:** No further D2 rail or local continuity sweep is needed. Seek a manufacturer 4P marking/package match consistent with the series-junction measurements; retain the explicit equivalent-circuit assumption if identity cannot be recovered. Do not fit or bridge R46.

## E03 - RF supply / Q8 / C6

**Known:** Q8 centre B2/native5 is GND via V113/V115. V114 is U2.T2/native2 GND. Outer Q8 pads are joined by the earlier low-resistance result. Candidate Q8 P-source/native2 is on filtered VSYS; P-gate/native3 is driven by U2 output4 through R8, while U2 uses VDD. C6 now parallels C5 on H\_RF\_VDD/GND, supported by user proposal and visible supply/ground copper in IMG\_2431/IMG\_2432.

**Uncertain:** Q8 source-supply path remains independently qualified. C6 wiring is photo-supported; its value/rating remain unknown. F-P=10 ohm supports a supply-related path but does not establish a direct copper join. Exact Q8 maker remains a candidate. Gate-drive headroom is unverified: a hypothetical source=6V and gate-high=4.5V gives VGS=-1.5V, not guaranteed off for the candidate threshold range. These are illustrative, not measured voltages.

**Next:** Do not repeat C6-to-C5 local continuity: the parallel connection is adopted from photos. Agent reviews Q8 source copper and the earlier F-P=10 ohm result before selecting any hidden-path check. After passive routing, measure VDD plus Q8 source and gate high/low during scanning, relative to PCB GND, to compute VGS. Waveforms require a scope if available; use PCB GND for the scope reference.

## I01 - Candidate identities / transistor pin maps

**Known:** Primary datasheets and marking evidence are recorded; custom symbols use the selected candidate pin tables. D3 measured rail connections fit the BAV99 clamp candidate; its identity is not newly proven.

**Uncertain:** Short markings do not uniquely prove manufacturers. Candidate choices and some B/E/C or MOSFET source/drain assignments remain qualified. Western alternatives are redesign options, not identities of fitted parts.

**Next:** Use electrical pad behavior and package/marketing evidence for specific candidates. Do not request sharper photos where the existing marking is already legible.

## I02 - LED / motor connector

**Known:** V127 reaches the photo-selected LED common pair; the two PIC LED drives are mapped. MX512H OUTA/OUTB map to the modeled motor connector. User confirms LED1 contains red and green emitters in a square 1.5 x 1.5 mm body; the schematic now says Red / Green.

**Uncertain:** Which R30/R31 (PIC22/PIC23) branch lights red versus green, exact die pairing/polarity and manufacturer pin numbering remain unconfirmed. J5 physical A/B lead polarity remains provisional.

**Next:** Only if channel identification is needed, use diode mode from each resistor-fed LED contact toward its grounded contact and record the emitted colour; do not repeat known board-routing checks. Match J5 wire positions to U5 outputs by continuity.

# Tuning and local endpoints

## E04 - Antenna / five tuning cells

**Known:** All five PIC control inputs are mapped. Seven capacitor-midpoint comparisons gave OL. User now confirms ANT2 directly to D2.L.

**Uncertain:** The model uses PNPs with emitter/native2 at GND and base driven through a resistor by nonnegative GPIO. This is not ordinary forward-operated PNP switching. Ground association, pad orientation/type or an RF junction-switching role may be wrong/incomplete. Capacitor-midpoint antenna continuations remain unresolved. ANT2-D2.L is now verified; further tuning/antenna connections remain open.

**Next:** Photo audit confirms the visible Q3 paired-right/R13 and single-pad/C10 joins; do not request them again. Next check one Q3 cell: directional diode tests of its physical paired/single pads to constrain PNP/NPN/base assignment. In-circuit results may be ambiguous and do not uniquely distinguish collector/emitter. Then resolve only the hidden V107-to-antenna continuation from front/rear evidence. Preserve all seven midpoint OL exclusions; no bulk via sweep.

## E05 - Button circuit

**Known:** R41=330 ohm from marking 331; user confirms its right/model1 end reaches J1 VPP/MCLR. Its left/model2 end joins R40 lower, C35 lower and S1 lower-right. S1 upper-right reaches C35 upper, previously tied to GND/V018. R40 opposite end remains ICSP\_DAT/PIC28.

**Uncertain:** S1 left-hand common-pair assignments and released/pressed behavior remain inferred. The reconstructed pressed path is VPP through 330 ohm to GND, with ICSP\_DAT reaching the switch node through 10k. Operating purpose is not established by the annotation.

**Next:** Only verify the switch contact behavior if required: released versus pressed continuity of opposite rows. Do not repeat the annotated C35/R40/R41 joins or R41-to-VPP destination. No broad resistor-to-rail sweep.

## E06 - Local-only PIC / discrete nodes

**Known:** Every PIC pad has a modeled local net. PIC2 reaches C25/V031; PIC14 reaches TP11; PIC18 reaches TP17. D3.S-to-R42 measured 1.2 ohm; exact resistor end not named, pad2 selected from photo. R42.1 already reaches TP4/PIC25/RB4. PIC21/RB0 now reaches TP16/Q2.S, separate from VREF; this PIR route is resolved. C26.1 reaches PIC3/RA1 and the existing R28/R29/TP7 net; C25.1 remains PIC2/RA0. Their model2 pads share GND. IMG\_2431 now establishes TP3/R6.2/R7.1/D1.S common drive and D1.R to U2 native1/C9.1/R7.2. PIC13-to-TP3 retains the previous user evidence.

**Uncertain:** Onward roles/connections of PIC2/RA0, PIC14/RC3 and PIC18/RC7 remain unknown. RF common drive and D1 local routing are now photo-supported, but actual excitation waveform and candidate diode/gate behavior remain unmeasured. H\_BAT\_SENSE continues through R4 to PIC24/RB3; its sensing role and H\_D3\_SIGNAL operating role remain unverified.

**Next:** Do not repeat visible TP3/R6/R7/D1 joins. Use original photos/vias to select hidden endpoints for remaining local-only PIC nodes; assess R3-R4-C28 together with PIC24. Defer RF timing to the powered session after tuning/device review.

# PIR, receiver and protection

## E07 - PIR interface / U6 / J3

**Known:** J3 pin1 supply, pin2-R38 K/right/model1 at 1 ohm and pin3-GND at 1 ohm. U7.1/2/3 map to J3.3/2/1. PIC21/RB0 reaches TP16/Q2.S; the former PIC21-VREF claim is withdrawn after TP16-VREF measured 0.8 Mohm. U6 pin5 externally joins pin2/VIN at 1 ohm; pin4 is intentionally unused. Candidate S-812C33AMC has internally NC pins4/5. Empty U6A now uses native reference U106: input pin3 after R37, output pin2 to J3.1, ground pin1. These native numbers supersede the old connector placeholder numbers.

**Uncertain:** Remaining U6 output-branch qualifications, actual PIR voltage, R33 branch and Q2 candidate pin functions remain qualified. Connector ordering and PIR output-to-controller destination are resolved.

**Next:** Do not repeat PIR-to-PIC routing tests. Agent checks remaining U6/R33 visible copper and candidate device orientation; only residual hidden ties or device behavior merit measurements.

## E08 - Receiver / bias / Q7

**Known:** U3.5 and R22.2 are VREF; R22.1 is GND. U3.6 reaches R23 B/left. R23 C/right, C40 lower and R42 upper are one measured junction (1 ohm). The old C40 detector-output link is withdrawn. Prior C18/TP6/VREF high-resistance exclusions remain; Q7 supply-switch topology is still inferred. v0.9.19 user annotated local copper: R47 left -> U3.1/R27 right; R47 right + C42 lower + R25 lower + R48 right share RX\_STAGE\_LINK. C42 upper -> GND inferred from continuous copper beside U3.4 in IMG\_2429/2434. R25-GND and R48-VREF old assumptions withdrawn. R47/C42 remain DNP. No meter reading supplied.

**Uncertain:** Remaining receiver bias/filter hypotheses, Q7 pin functions and operating transfer function. C23/R25 now continues into R48 through RX\_STAGE\_LINK; R42.1-TP4/PIC25 retains its independent evidence. H\_RX\_DETECT is the C23.2-R25.1 series junction; it continues through R25 into RX\_STAGE\_LINK/R48. Its historical name does not establish a detector output or justify adding another wire.

**Next:** IMG\_2429/2434 photo audit agrees with the existing local C23/R25/R47/R48 paths; no extra detector-to-MCU wire follows from the historical H\_RX\_DETECT name. Review the existing U3.6-R23-R42/TP4/PIC25 interface and Q7 supply mechanism before choosing a hidden endpoint or powered waveform. Do not repeat the completed local joins or rejected C40-to-output/R23-VREF/R25-GND/R48-VREF checks.

## E09 - Power input / Q1 / rail voltages

**Known:** BATTERY+ and VSYS are separate. User confirms Q1.L to U5.4 and both L1/L2 supply sides on VSYS. VDD TP is the separate logic rail. R1A matches FMOS3401A marking.

**Uncertain:** Exact Q1 maker and gate operation, unmeasured rail voltages, and U1 enable/option branches. Reverse-polarity protection is an interpretation, not a powered test.

**Next:** Cross-check Q1 gate/R44 and U1 enable wiring first. After routing reconciliation, use one powered session on the original hardware/supplies: BATTERY+, VSYS, VDD, VREF, U6 output, Q8 source/gates and both U5 control inputs, idle and during a triggered scan. Receiver power may be switched. No battery/PIC/RF redesign.

## E10 - ICSP protection / D6 / J1

**Known:** D4/D5=5U supports SD05 TVS candidates. D6=G3 has at least two documented Zener candidates. ICSP signal routes are photo-derived. J1 square-pad/VPP end is resolved from IMG\_2437 and corroborated by IMG\_2439: 1 VPP, 2 VDD, 3 GND, 4 DAT, 5 CLK. Electrical signal destinations are unchanged. B22-B27 resolve physical D6 upper V to GND and lower W to VPP, each 1 ohm. v0.9.20 B22-B27: upper photo contact V -> GND, lower W -> VPP, each 1 ohm. B29 red V/black W about 0.7V; B30 reverse OL. Supports A=V/GND (pin2), K=W/VPP (pin1). G3 identity and Zener breakdown remain unverified; no 3.9V measurement.

**Uncertain:** D6 exact identity/class, breakdown voltage and ICSP voltage compatibility remain unknown. The 3.9V value is only a marking candidate, not a measured clamp voltage. J1 geometry remains photo-estimated.

**Next:** No further D6 routing or polarity checks. Resolve actual breakdown/type only with appropriate part evidence or a controlled isolated test if required; diode mode does not measure breakdown.

# Values, mechanics and scope

## V01 - Unmarked passive values

**Known:** C5 is 470 uF / 16 V; 38 fitted resistors have nominal marked values. User LCR readings: L1 1.4 uH, L2 2.2 uH, now displayed in the schematic. R8/C11 measurements support 0603/1206 packages.

**Uncertain:** 44 fitted ceramic values remain unmeasured. L1/L2 test frequency, series/parallel mode, fixture compensation and in-circuit/isolation condition are unreported; nominal values, magnetic construction, tolerance, DCR and current ratings remain unidentified. Other voltage, dielectric and power ratings are separate gaps.

**Next:** Record L1/L2 measurement condition and test frequency if available; do not repeat readings yet. After one tuning cell is verified, prioritize RF/receiver capacitances. C11/C12 share both modeled nodes: an in-circuit reading is their combined value. Record frequency, mode and isolation; separate individual values only when needed.

## M01 - Package geometry / PCB reproduction

**Known:** 148/149 entries have candidate stock footprints; only LED1 remains blank. C5 is measured 6.33 mm diameter x 16 mm high and uses the nominal 6.3 mm radial footprint. S1 measures 6 x 6 x 4 mm plus 2 mm actuator (6 mm overall). LED1 is red/green, square 1.5 x 1.5 mm. R8/C11 bodies are measured; J1 numbering is corrected.

**Uncertain:** C5 lead pitch/drill fit remain inferred (stock 2.5 mm/0.8 mm); its stock 7 mm housing differs from the measured 16 mm. LED1 land pattern/numbering remain unresolved. S1 exact pad fit is unverified and stock height is 4.3 mm versus measured 6 mm. Other pitches, drills and board geometry remain partly photo-based.

**Next:** For PCB reproduction, verify C5 lead spacing and S1 pad fit; use measured C5 16 mm and S1 6 mm heights for clearance. Obtain a matching LED1 vendor footprint/pad drawing. No repeat body measurement is needed.

## D01 - Empty options

**Known:** All empty-option pads have modeled local nets. Rear C49 is an unpopulated ANT2-to-GND option. R46, U7, R47/C42 and J6/Q9/R35/R36 retain their user/photo-supported routes and DNP gaps. U6A is fully mapped: single pad after R37, upper-right PIR supply, lower-right GND. Options are drawn with their functional blocks; dated evidence remains in resolved\_subquestions.

**Uncertain:** J6.2/R36.1 onward role and intended option values remain unknown. Q9 NPN and U6A regulator types describe candidate roles for empty footprints. The user chose to retain the U6A regulator symbol and its U6.3/U106.2 ERC conflict. Native U106 mapping remains pin3=input, pin2=PIR supply, pin1=GND; the review example swapped pins1/3. Simultaneous regulator population is not validated.

**Next:** C49 routing is resolved; keep it as an unpopulated ANT2-to-GND option. Keep Q9/J6 with PIC outputs and U6A with the PIR regulator. Only hidden onward continuations and intended optional part choices remain to assess.

## F01 - Function / internal layers / daughterboard

**Known:** The one-sheet schematic has a native/model connectivity check and retained photo/via evidence; see the current validation counts.

**Uncertain:** No powered functional validation, actual layer stack or routed KiCad PCB has been created. The PIR daughterboard internal circuit is not reconstructed. MCU firmware behavior and operating timing are not established.

**Next:** Finish priority electrical gaps first. Then test defined functions/voltages and decide whether physical PCB reproduction and daughterboard reverse engineering are needed.

## Every unrecovered ceramic value

C1, C2, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43, C44, C45, C46, C47, C48

C5 is already 470 uF / 16 V. In-circuit LCR readings may include parallel components and semiconductor paths; record frequency, mode and whether a lead was isolated. Geometry cannot establish capacitance, dielectric or voltage rating.

## Closed issues retained as history

E02 - User identifies D3.L=GND and D3.R=VDD through V082. Move only D3.R off the hypothetical receiver supply. D3.S-to-R42 measures 1.2 ohm (resistor end not stated; photo selects pad2). Signal function remains with E06 and candidate identity with I01.

# PIC connections

Physical package pins. All have modeled local nets; the evidence column distinguishes confirmed endpoints from photo-derived or incomplete branches. Use the separate numbered photo guide for physical locations.

| Pin | Function | Local destination / evidence                                                             |
|-----|----------|------------------------------------------------------------------------------------------|
| 1   | RE3/MCLR | J1.1/VPP plus D6 cathode; D6 W-VPP 1 ohm, forward 0.7V from GND                          |
| 2   | RA0      | C25 signal pad / V031; onward destination unknown                                        |
| 3   | RA1      | C26 signal pad (user annotation); R28 + R29 + TP7 (earlier meter/trace)                  |
| 4   | RA2      | VREF TP via V035; user confirms via-to-TP; C39.1 local photo route                       |
| 5   | RA3      | V036-V108 to R14; user continuity report                                                 |
| 6   | RA4      | V037-V102 to R15; user continuity report                                                 |
| 7   | RA5      | V039-V097 to R16; user continuity report                                                 |
| 8   | VSS      | GND / V041 photo anchor                                                                  |
| 9   | RA7/OSC1 | Y1 oscillator; photo-derived                                                             |
| 10  | RA6/OSC2 | Y1 oscillator; photo-derived                                                             |
| 11  | RC0      | V047-V091 to R11; user continuity report                                                 |
| 12  | RC1      | R18.1 via V053-V070; user-confirmed; VREF link explicitly rejected                       |
| 13  | RC2      | TP3 via V049 (user); R10/R6/R7/D1 midpoint from IMG_2431; excitation waveform unmeasured |
| 14  | RC3      | TP11; visible route; onward destination unknown                                          |
| 15  | RC4      | V022-V016 to U5 pin2 / INA; user-confirmed                                               |
| 16  | RC5      | V023-V019 to U5 pin3 / INB; user-confirmed                                               |
| 17  | RC6      | R13; user meter / visible trace                                                          |
| 18  | RC7      | TP17; visible route; old R13 guess withdrawn                                             |
| 19  | VSS      | GND / C32 / V045 photo anchor                                                            |
| 20  | VDD      | V042 to board VDD TP user-confirmed; C32.1 local photo route                             |
| 21  | RB0      | TP16/Q2.S; user correction. Old VREF join rejected; TP16-VREF 800 kohm                   |
| 22  | RB1      | TP2 / R31; user meter / visible trace                                                    |
| 23  | RB2      | TP1 / R30; user meter / visible trace                                                    |
| 24  | RB3      | R4; photo-derived; sensing role remains inferred                                         |
| 25  | RB4      | TP4 / V017 / R42.1 / R44.1; user continuity; C41 photo-derived                           |
| 26  | RB5      | TP10 / R35; user meter / visible trace                                                   |
| 27  | RB6/CLK  | ICSP CLK / D4 / R32; photo-derived                                                       |
| 28  | RB7/DAT  | ICSP DAT / D5 / R40; photo-derived                                                       |

## Every remaining open pad

**Populated entries:** None.

**Empty options:** None.

TP9 was withdrawn as an unsupported inherited inventory entry; rear C49 is photographed and now confirmed between ANT2 and GND. Both pads are resolved; its DNP classification is retained. U6 pin4 is unused by user/photo evidence; pin5 externally joins pin2/VIN (1 ohm), although internally NC in the candidate datasheet. Photo pad S means single pad, not necessarily source. The completion checklist includes closure criteria and the full component audit. Sources: reconstruction.json, remaining\_work.json, via\_audit.json, native validation and preserved measurement records.